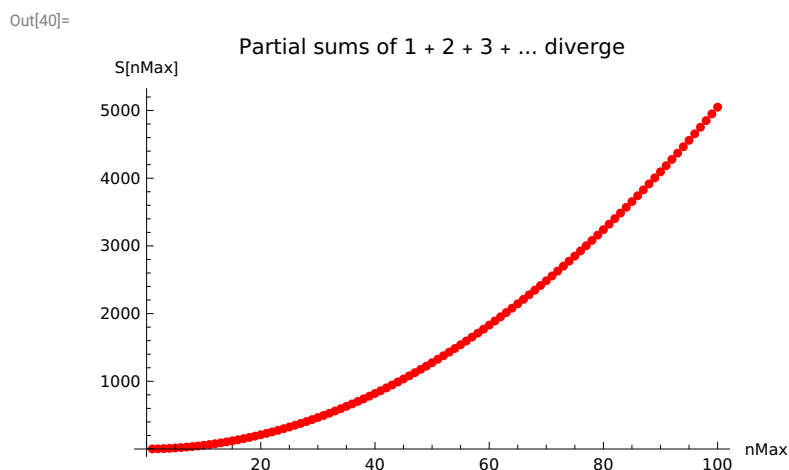


Vacuum Energy and the Casimir Effect: A 1+1D Toy Model

The Casimir effect is a measurable quantum mechanical force arising from the zero-point energy of the vacuum. In this notebook I explore a simplified 1+1 dimensional toy model, starting from the problem of regularising a divergent sum and, then moving through a closed-form expression for the force between two plates, and then finally finding a finite, physical force between two plates.

Naive Sum of $1 + 2 + 3 + \dots$ Divergence Representation

```
In[39]:= partialSums = Table[Sum[n, {n, 1, nMax}], {nMax, 1, 100}];  
ListPlot[partialSums,  
  PlotLabel -> "Partial sums of 1 + 2 + 3 + ... diverge",  
  AxesLabel -> {"nMax", "S[nMax]"},  
  PlotStyle -> {Red, PointSize[0.015]}]
```



The vacuum energy of a simple 1D system, summed mode by mode, looks like the sum from 1 to Nmax. As Nmax increases, it grows to no bound, and cannot represent a physical and measurable energy.



Now, we introduce a regulator $k > 0$ that suppresses high-frequency modes, making the sum convergent.

We define the function:

$$T(n_{max}, k) = \sum_{n=1}^{n_{max}} n e^{-k n} \cos(k n).$$

The regulator is removed by taking $k \rightarrow 0$ and then we recover the original sum.

```
In[41]:= (* Regularised sum *)
t[n_, k_] := n Exp[-k n] Cos[k n]
T[nMax_, k_] := Total[Table[t[n, k], {n, 1, nMax}]]

(* Show that as k→0, t[n,k] → n *)
Limit[t[n, k], k → 0]
```

Out[43]=

n

```

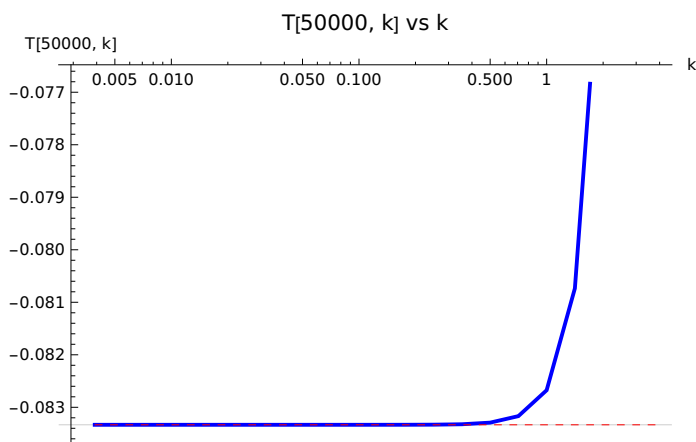
In[44]:= (* Generate k values logarithmically spaced *)
kValues = NestList[N[##/Sqrt[2], 20]&, N[2^2, 20], 20];

(* Compute T[nMax, k] for large nMax *)
data = Table[{k, N[T[50000, k], 10]}, {k, kValues}];

ListLogLinearPlot[data, Joined → True,
  PlotLabel → "T[50000, k] vs k",
  AxesLabel → {"k", "T[50000, k]"},
  PlotStyle → Blue,
  GridLines → {{}, {-1/12}},
  Epilog → {Dashed, Red,
    Line[{{Log[Min[kValues]], -1/12}, {Log[Max[kValues]], -1/12}}]}]

```

Out[46]=



As $k \rightarrow 0$, the regularised sum converges to $-1/12$, shown by the red dashed line.

This is the well known result of the identity $1+2+3+\dots=-1/12$, which is not a statement about ordinary summation but about the finite part that survived regularisation.

Same logic is used for physical Casimir energy calculation below:



Consider the physical setup: two plates separated by distance d in a $1+1$

dimensional toy model. The vacuum energy between them is regularised similarly as k , using an ultraviolet cutoff a :

$$E(d, a) = -\frac{1}{2} \frac{\partial}{\partial a} \sum_{n=1}^{\infty} e^{-a n \pi/d}.$$

The sum is a simple geometric series, so it can be evaluated in closed form before differentiated:

```
In[47]:= (* The sum a geometric series, which Mathematica evaluates symbolically then the close
VacuumE[d_, a_] := -1/2 D[Sum[Exp[-a Pi n/d], {n, 1, Infinity}], a]
```

```
VacuumE[d, a]
```

```
Out[48]=
```

$$\frac{e^{\frac{a \pi}{d}} \pi}{2 d \left(-1 + e^{\frac{a \pi}{d}} \right)^2}$$

```
In[49]:= (*expand around a = 0 to separate the divergent part
from the finite, physical part *)
series = Series[VacuumE[d, a], {a, 0, 0}]
(* Output: contains a 1/a^2 term (divergent as a → 0) plus a finite a^0
term plus higher-order terms in a *)

finitePart = SeriesCoefficient[VacuumE[d, a], {a, 0, 0}]
(* Output: -Pi/(24 d) - the coefficient of a^0 specifically, pulled out
directly from the full series *)
```

```
Out[49]=
```

$$\frac{d}{2 \pi a^2} - \frac{\pi}{24 d} + O[a]^1$$

```
Out[50]=
```

$$-\frac{\pi}{24 d}$$

This closed-form expression contains an unphysical regulator a . To extract the physical content, we expanded around $a = 0$ and isolate the finite piece. The

divergent $1/a^2$ term represents the self-energy of each plate and does not depend on the separation d , so it carries no information about the interaction between the plates and is discarded)



The total vacuum energy of the full system - between the plates and beyond them, over a total system length L much greater than d - is given by:

$$E_{tot}(d, a, L) = E(d, a) + E(L - d, a)$$

The force between the plates is $F = -\frac{d E_{tot}}{d d}$, evaluated in the physical limit where the UV regulator a is removed ($a \rightarrow 0$) and the system size is taken to infinity ($L \rightarrow \text{Infinity}$), in that order.

In[51]:

```

Etot[d_, a_, L_] := VacuumE[d, a] + VacuumE[L - d, a]

(* Differentiate with respect to a dummy variable dvar rather than d
   itself. This keeps the differentiation symbolic and safe even when
   Force[d] is later called with specific numeric values of d (e.g.
   by Plot) *)
forceSymbolic = D[Etot[dvar, a, L], dvar];

(* The two Limit calls are nested deliberately and in this specific
   order: remove the UV cutoff first (a → 0), then send the system
   size to infinity (L → Infinity) *)
forceLimited = Limit[Limit[forceSymbolic, a → 0], L → Infinity];
Force[d_] := forceLimited /. dvar → d

Force[d]
(* Output: Pi/(24 d^2) - a finite, d-dependent force *)

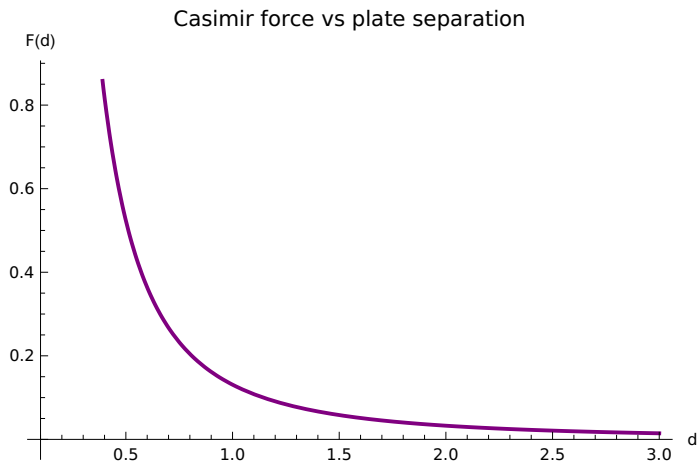
Plot[Force[d], {d, 0.1, 3},
  PlotLabel → "Casimir force vs plate separation",
  AxesLabel → {"d", "F(d)"},
  PlotStyle → {Thick, Purple}]
(* Output: a smooth curve falling off as 1/d^2, confirming the plates
   are pulled together more strongly at small separations *)

```

Out[55]=

$$\frac{\pi}{24 d^2}$$

Out[56]=



The resulting force is attractive and falls off as $1/d^2$. Physically, this arises because the plates restrict which vacuum field modes can exist between them, modifying the local mode spectrum relative to free space and producing a net inward pressure. This is the Casimir force in this simplified 1+1 dimensional model.



Comparison to the Real 3D Casimir Effect

The real Casimir force between two parallel conducting plates is given by:

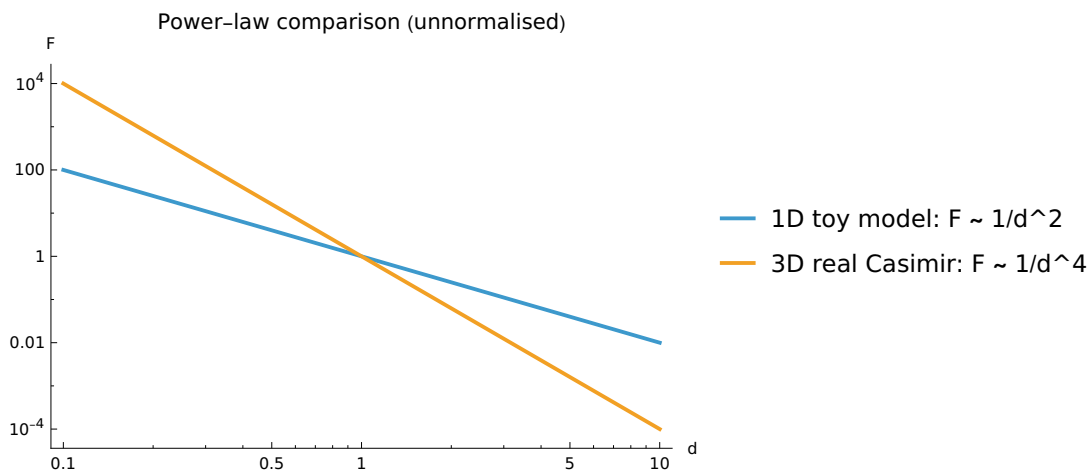
$$F_{3D}(d) = -\frac{\pi^2 c}{240 d^4}$$

Both results share the same essential physics - zero-point energy regularisation produce an attractive force - but are different in power law due to dimensionality

In[57]:=

```
LogLogPlot[{1/d^2, 1/d^4}, {d, 0.1, 10},
  PlotLegends -> {"1D toy model: F ~ 1/d^2", "3D real Casimir: F ~ 1/d^4"},
  PlotLabel -> "Power-law comparison (unnormalised)",
  AxesLabel -> {"d", "F"}]
```

Out[57]=



The steeper $1/d^4$ line in 3D reflects the larger number of field modes available for the regulator to act on in higher dimensions - more transverse directions mean more modes contribute to the mode-counting argument, which changes the power law even though the underlying regularisation logic is identical. This 1+1D toy model captures the essential mechanism - divergent sum, regularise, extract finite part, take the physical limit - that underlies the genuine, experimentally measured Casimir force.